

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 1: VFR Flight Planning & Navigational Calculations

Syllabus Source: CAE Oxford (pp. 1-40) | Pass Target: $\geq 90\%$ AviationExam

1. THE NAVIGATIONAL VECTOR TRIANGLE & TERMINOLOGY

Navigation is the geometric combination of aircraft heading/airspeed with the wind vector:

- True Track (TR): The actual path over the earth's surface that the aircraft intends to follow or is tracking.
- True Airspeed (TAS): The calibrated airspeed corrected for altitude and temperature. Represents speed through undisturbed air.
- Wind Vector (W/V): Given as direction (degrees True) FROM WHICH the wind blows, and speed (knots).
- Drift Angle & Wind Correction Angle (WCA): The angle between True Heading and True Track. Drift is Right (+) if wind pushes aircraft right of track. WCA opposes drift: Heading = Track - (+- Drift).
- Groundspeed (GS): The actual speed of the aircraft across the ground: $GS = TAS \pm \text{Wind Component}$.
- Conversion Chain: True Heading (TH) $\pm$ Magnetic Variation = Magnetic Heading (MH) $\pm$ Compass Deviation = Compass Heading (CH) ('Variation West, Magnetic Best; Variation East, Magnetic Least').

2. ETE & FUEL CONSUMPTION FORMULAS

- Estimated Time En Route (ETE): Time (hours) = Distance (NM) / Groundspeed (kt). Time (minutes) = (Distance / GS) x 60.
- Leg Fuel Burn: Fuel (liters or kg) = Time (hours) x Fuel Consumption Rate (liters/hr or kg/hr).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Dead Reckoning Wind Triangle &

Problem: True Track = 090, TAS = 150 kt. Wind = 360/30 kt. Magnetic Variation = 5 W.

Compass Deviation = +2. Step 1: Solve Wind Triangle: - Wind is 90 degrees from left

(360). $\sin(WCA) = (V_{\text{wind}} / TAS) \times \sin(\text{Angle}) = (30 / 150) \times 1.0 = 0.20$. - WCA = -12

degrees (drift to right -> steer left). True Heading TH = 090 - 12 = 078. Step 2:

Magnetic Variation: West is Best -> ADD: Magnetic Heading MH = 078 + 5 = 083. Step 3:

Compass Deviation: Compass Heading CH = 083 - (+2) = 081. Step 4: Groundspeed GS =

$\sqrt{150^2 - 30^2} = 147$ knots.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: VFR Leg Heading, Groundspeed, ETE & Fuel

SCENARIO (Standard Navigation Flight Planning Drill): A pilot plans a cross-country leg: - True Track = 090° - Distance = 160 NM - True Airspeed (TAS) = 120 knots - Forecast Wind = 360° at 30 knots (Direct crosswind from the Left!) - Magnetic Variation = 5° West - Fuel Flow = 40 liters/hour QUESTION: Calculate (1) Wind Correction Angle & True Heading, (2) Magnetic Heading, (3) Groundspeed, (4) ETE, (5) Fuel Burned. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Wind Correction Angle (WCA): - Wind is from 360° against a Track of 090° (Wind angle = 90° direct from Left). - Crosswind component = $30 \text{ kt} \times \sin(90^\circ) = 30 \text{ knots}$. - Headwind component = $30 \text{ kt} \times \cos(90^\circ) = 0 \text{ knots}$. - Rule of Thumb: Drift (degrees) ~ (Crosswind kt / TAS kt) x 60 - Drift = $(30 / 120) \times 60 = 0.25 \times 60 = 15^\circ$ to the RIGHT. - To counter a 15° Right drift, you must steer 15° to the LEFT (-15° WCA)! - True Heading (TH) = Track - Drift = $090^\circ - 15^\circ = 075^\circ$ True! Step 2: Calculate Magnetic Heading (MH) ('Variation West, Magnetic Best' -> ADD!): - MH = True Heading + Var West = $075^\circ + 5^\circ = 080^\circ$ Magnetic! Step 3: Calculate Groundspeed (GS) using Pythagoras Theorem on the vector triangle: - GS = $\sqrt{\text{TAS}^2 - \text{Crosswind}^2}$ - Headwind - GS = $\sqrt{120^2 - 30^2} = 0 = \sqrt{14,400 - 900} = \sqrt{13,500} = 116.2$ knots. Step 4: Calculate Estimated Time En Route (ETE): - Time = (Distance / GS) x 60 = $(160 \text{ NM} / 116.2 \text{ kt}) \times 60 = 1.377 \text{ hours} \times 60 = 82.6 \text{ minutes}$ (1h 23 min). Step 5: Calculate Leg Fuel Burn: - Fuel = Time (hours) x Fuel Flow = $1.377 \text{ hours} \times 40 \text{ L/hr} = 55.1 \text{ liters}$. FINAL ANSWER: True Heading = 075°; Magnetic Heading = 080°; GS = 116 kt; ETE = 83 minutes; Fuel = 55 liters.

[!] EXAM TRAP (AVIATIONEXAM) - Variation East vs West Memory Aid

- 'Variation West, Magnetic Best' -> Magnetic is BIGGER (ADD West variation to True Heading). - 'Variation East, Magnetic Least' -> Magnetic is SMALLER (SUBTRACT East variation from True Heading).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If True Track is 180°, True Airspeed is 150 kt, wind is 090°/30 kt, and variation is 10° East, what is the MAGNETIC HEADING?

- [A]: 160°
- [B]: 170°
- [C]: 190°
- [D]: 180°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Wind from 090° pushes aircraft to the Right (Drift +). Crosswind = 30 kt. Drift = $(30/150) \times 60 = 12^\circ$ Right. True Heading = $180^\circ - 12^\circ = 168^\circ$ True. Magnetic Heading = $168^\circ - 10^\circ$ East (East is least!) = 158° (~160°).

Q2: What is the effect of a 90° pure crosswind on aircraft Groundspeed compared to True Airspeed?

- [A]: Groundspeed equals TAS
- [B]: Groundspeed is SLIGHTLY LESS than TAS, because aircraft nose must angle into the wind
- [C]: Groundspeed is higher than TAS
- [D]: Groundspeed drops to zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Because the aircraft points into the wind to maintain track (WCA), a portion of its forward TAS vector is angled sideways, reducing the forward groundspeed vector: $GS = \sqrt{TAS^2 - XW^2}$.

Q3: In navigational flight planning, how is 'DRIFT' defined?

- [A]: The angle between True North and Magnetic North
- [B]: The angular difference between the Heading (aircraft longitudinal axis) and the Track (actual flight path over the ground)
- [C]: The climb angle
- [D]: The compass error

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Drift is the angle between heading and track caused by crosswind pushing the aircraft sideways.